

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Lecture 3 Animations: Student Guide

KKKC6353 Nanoelectronic Devices — learn by experimenting

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How to use this guide

Six interactive animations go with the Lecture 3 notes. They are at **faizverse.github.io/nanoelectronic-devices/L3** and work on a laptop or phone, online or saved for offline use.

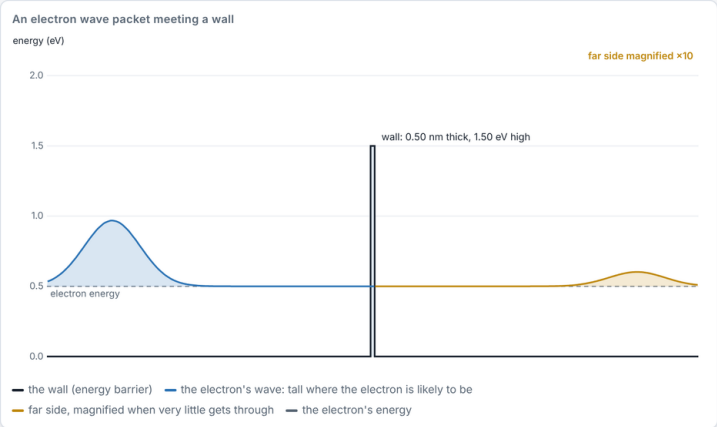
For each animation:

1. **Read first.** Skim the notes section listed, then read *What it shows* below. The page opens in a simple view; press **More detail** when you want the numbers and graphs.
2. **Predict, try, explain.** Before you move a slider, write down what you expect. Then try it. If the result surprised you, work out why.
3. **Answer the questions** at the end of each page in this guide, then check your answers under *Check yourself* on the web page.
4. **Note the big idea** in one sentence of your own.

Allow about 10 minutes per animation. Work through them in order: each builds on the one before.

#	Animation	Read first	Link
1	Tunnelling through a wall	Doc 1, § 1–3	tunnelling
2	Scanning tunnelling microscope	Doc 1, § 5	scanning-tunnelling-microscope
3	Tunnel diode	Doc 2, § 1–3	tunnel-diode
4	Resonant-tunnelling diode	Doc 2, § 4	resonant-tunnelling-diode
5	Tunnel FET versus MOSFET	Doc 2, § 5	tunnel-fet
6	Single-electron transistor	Doc 3, § 1–3	single-electron-transistor

1 Tunnelling through a wall



Link: https://faizverse.github.io/nanoelectronic-devices/L3/tunnelling.html
Read first: Document 1, Sections 1–3

What it shows. Send an electron, as a wave packet, at a wall it does not have the energy to climb. Most of it bounces back, but a little appears on the far side. Make the wall thicker or higher and watch how fast the chance falls.

A wave, not a ball. The blue hump is the electron, drawn as a wave packet: it is tall where the electron is likely to be. It moves towards a wall that is higher than its energy (the grey line). A ball would simply bounce back.

Most bounces, some gets through. When the packet hits the wall, most of it reflects. A smaller packet appears on the far side and carries on with the same energy. The size of that packet is the chance that the electron tunnelled through. When the chance is tiny, the far side is drawn magnified (gold) so you can still see it.

The wall must be thin. Make the wall a little thicker and the far-side packet shrinks enormously: the chance falls by the same factor for every extra slice of wall. Raising the wall, or using a heavier particle, does the same.

Above the wall. Give the electron more energy than the wall and most of it goes over, yet a little still reflects, which a ball would never do.

> **The big idea.** An electron can pass through a wall it does not have the energy to climb, because it is a wave. The chance falls by the same large factor for every extra slice of wall.

Try this

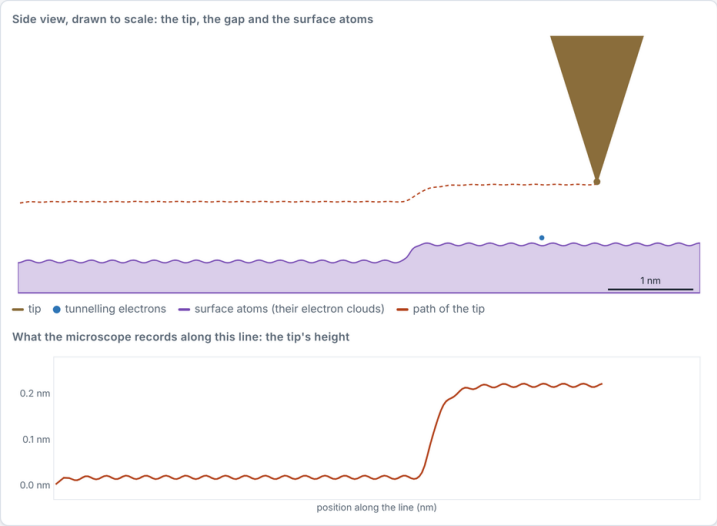
1. Default wall (0.5 nm, 1 eV above the electron): read the chance of getting through. **Predict** the chance at 1 nm, then check.
2. Back at 0.5 nm, raise the wall height so it is 2 eV above the electron. Which matters more, doubling the thickness or doubling the height?
3. Switch to an electron in GaAs. Does more or less get through? Why?
4. Set the electron's energy above the top of the wall. Does all of it get through?

Questions (check your answers on the page, under *Check yourself*)

- Does the electron lose energy by tunnelling?

- Why does doubling the thickness do so much more than doubling the height?
- Why do electrons tunnel in devices but atoms do not?

2 Scanning tunnelling microscope



Link: https://faizverse.github.io/nanoelectronic-devices/L3/scanning-tunnelling-microscope.html
Read first: Document 1, Section 5

What it shows. A sharp tip scans a nanometre above a surface while electrons tunnel across the gap. Because the current changes about ten times for every tenth of a nanometre, the tip can trace single atoms. Try both scanning modes, and crash the tip into a step.

A gap electrons cannot cross, except by tunnelling. The tip hovers about a nanometre above the surface. Electrons tunnel across the gap (blue dots), giving a current of about a nanoampere.

Extremely sensitive to distance. The tunnelling current falls by about ten times for every tenth of a nanometre the gap widens. So the current "feels" every bump of the surface, even single atoms only a few hundredths of a nanometre high.

Constant current (the usual mode): a feedback circuit moves the tip up and down to keep the current fixed, so the tip follows the bumps. Its height is the image. **Constant height**: the tip stays level and the current itself is recorded. It is faster, but the tip can crash into a step.

Scan too fast and the feedback cannot keep up: the image smears.

> **The big idea.** The tunnelling current changes about ten times for every tenth of a nanometre, so a tip held a nanometre above a surface can trace single atoms.

Try this

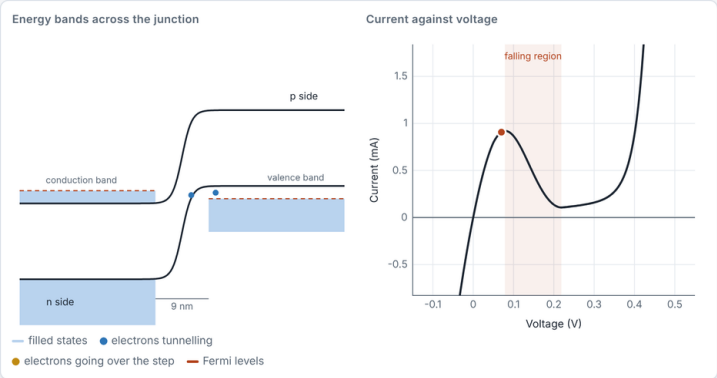
1. Constant current: watch the tip follow each atom. How high are the bumps it traces?
2. Open More detail. Change the gap from 0.7 to 0.6 nm. By how much does the current change?
3. Surface with a step, constant height: how much does the current jump at the step? Now set the gap to 0.45 nm and try again.

4. In More detail, let the image build up for the surface with a missing and an extra atom. Find both.

Questions (check your answers on the page, under *Check yourself*)

- Why can an STM see bumps only a few hundredths of a nanometre high?
- Why is constant-current mode safer than constant-height mode?
- Why does an STM need a conducting surface?

3 Tunnel diode



Link: https://faizverse.github.io/nanoelectronic-devices/L3/tunnel-diode.html
Read first: Document 2, Sections 1–3

What it shows. A p–n junction doped so heavily that electrons can tunnel straight across it. Slide the voltage and watch the filled states on one side line up with the empty states on the other, then slide past them: the current rises, falls, and rises again.

Heavy doping, thin wall. Both sides are doped so heavily that the junction is only about 10 nm wide, and the Fermi levels (red dashes) sit inside the bands: the n side has a strip of filled states in its conduction band, and the p side a strip of empty states at the top of its valence band.

Filled must face empty. An electron can only tunnel into an empty state at the same energy. A small forward voltage lifts the n side's filled strip opposite the p side's empty strip, and electrons tunnel across (blue): the current rises to a **peak**.

Then it falls. More voltage slides the filled strip past the empty one, until it faces only the band gap, where there are no states. Tunnelling stops and the current falls: **negative differential resistance**. At larger voltages, electrons go over the step as in an ordinary diode (gold), and the current rises again.

Reverse. With reverse voltage, the p side's filled valence band faces the n side's empty conduction band, and a large current flows straight away.

> **The big idea.** Electrons can only tunnel into empty states at the same energy. As the voltage rises, the filled states line up with empty ones and then slide past them, so the current rises, falls and rises again.

Try this

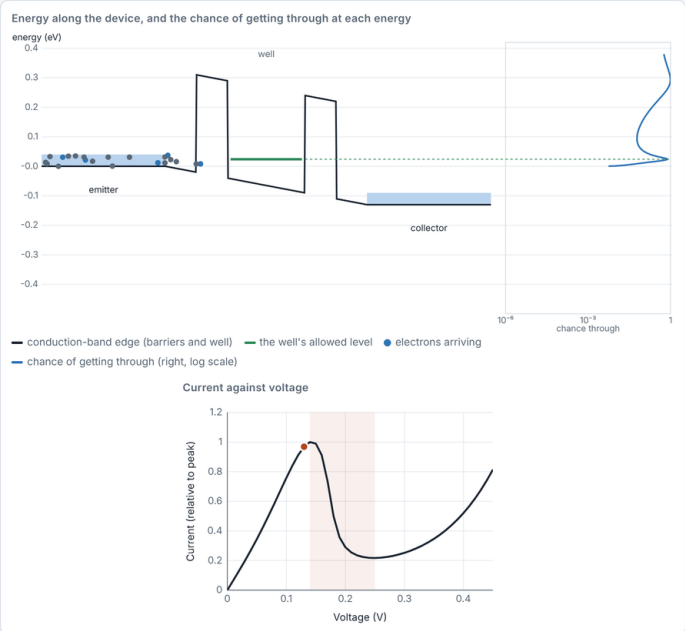
1. Raise the voltage slowly from zero and stop at the peak. Which states face each other?

2. Keep going to the valley. What lies opposite the n side's filled strip now?
3. **Predict** what lower doping does to the peak. Check, and look at the junction width.
4. Switch to gallium arsenide. Why does the ordinary diode current start at a higher voltage?

Questions (check your answers on the page, under *Check yourself*)

- Why does the current fall between the peak and the valley?
- Why does lower doping give a much smaller peak?
- Is the diode a source of power in its falling region?

4 Resonant-tunnelling diode



Link: https://faizverse.github.io/nanoelectronic-devices/L3/resonant-tunnelling-diode.html

Read first: Document 2, Section 4

What it shows. Two thin walls with a quantum well between them let electrons through almost perfectly at one energy and hardly at all at others. Raise the voltage and watch the well's level line up with the incoming electrons, then drop below them.

A sharp filter. Two AlGaAs barriers a couple of nanometres thick enclose a GaAs quantum well with one allowed level (green). Electrons whose energy matches that level pass through both barriers almost every time; all others are almost always reflected. The blue curve on the right shows this: near 100% at the level's energy, tiny elsewhere.

The voltage moves the level. The voltage tilts the picture and pulls the level down. When it lines up with the electrons waiting in the emitter (left), they pour through: the current peaks. When it drops below the bottom of the emitter's band, no electron has the right energy and the current falls. That fall is negative differential resistance.

Watch the electrons. Electrons arrive at many energies. Those at the right energy go straight through; the rest bounce back.

> **The big idea.** Two thin walls with a quantum well between them let electrons through only at the well's level. The voltage pulls the level down past the incoming electrons, so the current peaks and then falls.

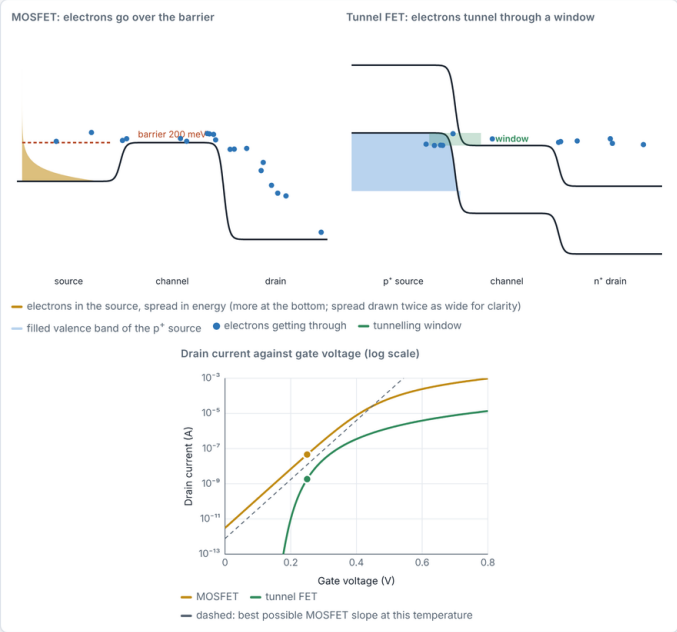
Try this

1. Raise the voltage to the peak. Where is the green level compared with the electrons on the left?
2. Go past the peak. Why does the current fall?
3. **Predict**: will a narrower well move the peak to a higher or a lower voltage? Check.
4. Make the barriers thicker. What happens to the blue transmission peak and to the current?

Questions (check your answers on the page, under *Check yourself*)

- How can an electron cross two barriers almost perfectly, when it rarely crosses one?
- Why does a narrower well move the peak to a higher voltage?
- Why is a resonant-tunnelling diode so fast?

5 Tunnel FET versus MOSFET



Link: https://faizverse.github.io/nanoelectronic-devices/L3/tunnel-fet.html
Read first: Document 2, Section 5

What it shows. An ordinary transistor switches by letting hot electrons over a barrier, so it can never switch more sharply than about 60 mV per tenfold change in current at room temperature. A tunnel FET opens a tunnelling window instead. Move the gate and the temperature and compare the two.

MOSFET (left). Electrons in the source have a spread of energies: most are low, a few are "hot". The gate raises and lowers a barrier in the channel. Only electrons above the barrier get through, and their number falls off smoothly with energy. That is why the current can change by at most ten times for every 60 mV on the gate at room temperature, and less sharply still when warmer.

Tunnel FET (right). The source is p-type, so its electrons sit in a filled valence band with a sharp top edge. With the gate off, the channel's conduction band is above that edge and nothing can tunnel. The gate pulls it down; once it drops below the edge, a window (green) opens and electrons tunnel straight through. The hot tail is cut off, so the current can switch on more steeply.

The catch. Tunnelling lets fewer electrons through than going over a barrier, so the tunnel FET's full on-current is much smaller.

> **The big idea.** A MOSFET lets hot electrons over a barrier, so it needs at least 60 mV per tenfold change in current at room temperature. A tunnel FET opens a tunnelling window instead and can switch more steeply, but carries less current.

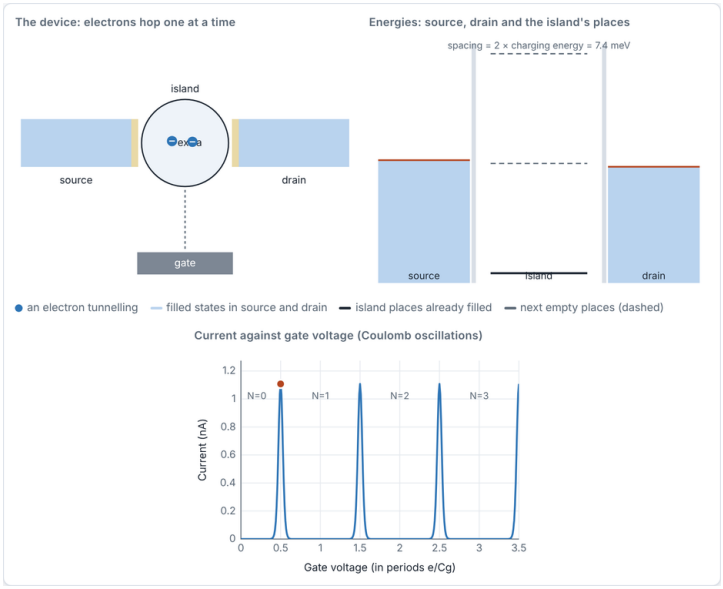
Try this

1. Sweep the gate from 0 to 0.8 V. Where does each device turn on?
2. At low current, read the steepness of each. Which beats 60 mV per decade?
3. **Predict** what heating to 400 K does to each device. Check.
4. Compare the two currents at 0.8 V. Which would make the faster circuit?

Questions (check your answers on the page, under *Check yourself*)

- Why can a MOSFET not beat 60 mV per decade at room temperature?
- Why is the tunnel FET's switching almost independent of temperature?
- Why are tunnel FETs not in today's chips?

6 Single-electron transistor



Link: https://faizverse.github.io/nanoelectronic-devices/L3/single-electron-transistor.html

Read first: Document 3, Sections 1–3

What it shows. Electrons tunnel onto a tiny island and off again, one at a time. Adding one more costs the charging energy, so the current is blocked until the gate lines up the next place on the island. Turn the gate, warm the device up, and shrink the island.

A tiny island. The island is joined to the source and drain only through two thin tunnel junctions. The gate is close to it but no electrons cross to the gate.

One more electron costs energy. Electrons repel, so adding one more to a small island costs the charging energy. On the right, the island's places form a ladder: filled places (solid) and the next empty ones (dashed). An electron can hop on from the source only if the next empty place is below the source's level, and off to the drain only if that place is above the drain's level.

The gate moves the ladder. Turning up the gate lowers the island's places. When the next empty place sits between the source and drain levels, electrons pass one by one: a current peak. Move on and the place fills and stays filled: blocked again, with one more electron on the island. So the current rises and falls repeatedly: **Coulomb oscillations**.

Heat and size. When the thermal energy approaches the charging energy, random kicks let electrons through anyway and the peaks wash out. A smaller island has a larger charging energy and survives higher temperatures.

> **The big idea.** Adding one electron to a tiny island costs the charging energy, so current is blocked except when the gate lines up the island's next empty place between source and drain. The current peaks once for every electron added.

Try this

1. Start where the current is blocked. Raise the gate slowly until current flows. Where is the next empty place on the ladder?
2. Keep raising the gate through two peaks. How many extra electrons are on the island after each?
3. **Predict** what warming the 100 nm island from 1 K to 30 K will do. Check.
4. Shrink the island to 2 nm and set 300 K. Are the oscillations still visible? Why?

Questions (check your answers on the page, under *Check yourself*)

- What blocks the current in Coulomb blockade?
- Why are the current peaks evenly spaced in gate voltage?
- Why does a room-temperature SET need an island of only a nanometre or two?